DS3D-HVAC Simulation Interface Specification

Official Communication Protocol, Telemetry Formats & Variable Dictionary for LabVIEW

Document Version:	1.1.0 (Integration Ready)	**Target Project:**	DS3D-HVAC Simulation
Transport Protocol:	UDP (User Datagram Protocol)	**Character Encoding:**	UTF-8
Unity Listen Port:	`9000` (LabVIEW TX → Unity)	**LabVIEW Listen Port:**	`5005` (Unity TX → LabVIEW)
Excel Reference:	`DS3D_HVAC_Variable_List.xlsx` (Tabs: *HVAC_Variables, Inbound_Commands, Outbound_Telemetry, Fault_Catalog*)		

1. Network Port Mapping & Architecture

Direction	Sender	Receiver	Port	Payload Category
Inbound	LabVIEW Dashboard	Unity Simulator	**9000**	Fault Trigger Commands (101–114), Reset, Skip, Sliders/Animation Telemetry
Outbound	Unity Simulator	LabVIEW Dashboard	**5005**	Step Accomplished Events, Wrong Attempt Counters, Fault Completion Summary

Automatic LAN IP Discovery

When running across two separate machines over Ethernet or Wi-Fi, Unity automatically updates its outbound target IP to match the IP of the incoming LabVIEW packet.

2. Inbound Formats (LabVIEW → Unity, Port 9000)

A. Trigger Procedural Fault (Standard JSON — Recommended)

```
{
  "command": "trigger_fault",
  "fault_code": 101
}
```

Shorthand JSON: `{"fault": 101}` | *Raw string fallback:* `101`

Behavior: Loads the fault procedure, highlights required tool for Step 1, frames camera, and resets any prior fault cleanly.

B. Reset Simulator Command

```
{
  "command": "reset"
}
```

(Raw text alternative: `reset`*)* — Returns simulator to idle 3D inspection mode.

C. Skip Step Command

```
{
  "command": "skip_step"
}
```

(Raw text alternative: `skip`*)* — Advances the active step during training mode.

3. Registered HVAC Fault Scenarios Catalog (Scene HVAC.unity)

Code	Subsystem	Fault Title & Diagnostic Scope	Step Count
`101`	Refrigeration Circuit	No HVAC operations	14 Steps
`102`	Blower & Ventilation	No ventilation	22 Steps
`103`	Ventilation Circuit	Reduced ventilation	15 Steps
`104`	Compressor / Refrigerant	Suction low and discharge low (Circuit 1)	15 Steps
`105`	Expansion / Pressure	Suction low and discharge low (Circuit 2)	14 Steps
`106`	Pressure Differential	High suction and high discharge	26 Steps
`107`	Compressor Unit	Compressor not operating	40 Steps
`108`	Cooling Capacity	Reduced cooling	33 Steps
`109`	Heating Subsystem	No Heating	15 Steps
`110`	Thermal Regulation	Over Cooling	8 Steps
`111`	Thermal Regulation	Over Heating	7 Steps
`112`	Emergency Systems	No emergency ventilation	14 Steps
`113`	Safety / Detection	Smoke identified in saloon	13 Steps
`114`	Mechanical Diagnostics	Compressor excessively noisy | Condenser Fan or Supply Blower	36 Steps

4. Outbound Telemetry (Unity → LabVIEW, Port 5005)

Attempt Counting & Network Quietness Rules

1. **Wrong Tool Interaction:**
   - When a user selects an incorrect tool or interacts with an invalid target, **no message is sent over the network**.
   - Unity logs the attempt and increments the internal `wrong_attempts` counter for that step.
2. **Correct Tool Interaction & Step Accomplishment:**
   - When the user interacts with the correct tool and the step completes, Unity immediately sends the `step_completed` JSON packet.
   - The packet reports the accomplished step number and the exact count of wrong attempts made prior to succeeding.
3. **Fault Completion:**
   - When all steps in the fault are finished, Unity sends the `fault_completed` summary packet.

Payload 1: Step Accomplished (`step_completed`)

Sent upon successful completion of each step in the fault procedure:

```
{
  "event": "step_completed",
  "fault_code": 101,
  "step_number": 1,
  "wrong_attempts": 0,
  "total_steps": 14,
  "status": "success",
  "is_fault_completed": false
}
```

Example: Step 2 completed after 2 wrong tool attempts:

```
{
  "event": "step_completed",
  "fault_code": 101,
  "step_number": 2,
  "wrong_attempts": 2,
  "total_steps": 14,
  "status": "success",
  "is_fault_completed": false
}
```

Field Name	JSON Type	LabVIEW Cluster Type	Description
`event`	String	String	Constant value `"step_completed"`
`fault_code`	Number (int)	Numeric (I32)	The active fault code (e.g. `101`)
`step_number`	Number (int)	Numeric (I32)	1-based index of accomplished step (`1` , `2` , `3` ...)
`wrong_attempts`	Number (int)	Numeric (I32)	Count of incorrect tool attempts made before completing this step
`total_steps`	Number (int)	Numeric (I32)	Total steps in this fault scenario

Field Name	JSON Type	LabVIEW Cluster Type	Description
`status`	String	String	Execution outcome (always `"success"`)
`is_fault_completed`	Boolean	Boolean	`true` if this was the last step; otherwise `false`

Payload 2: Full Fault Completed Summary (`fault_completed`)

Sent when the final step is completed (or submitted) to summarize the entire performance:

```
{
  "event": "fault_completed",
  "fault_code": 101,
  "status": "success",
  "total_steps": 14,
  "wrong_attempts_per_step": [0, 2, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0],
  "is_fault_completed": true
}
```

Field Name	JSON Type	LabVIEW Cluster Type	Description
`event`	String	String	Constant value `"fault_completed"`
`fault_code`	Number (int)	Numeric (I32)	The completed fault scenario code
`status`	String	String	`"success"`
`total_steps`	Number (int)	Numeric (I32)	Total count of steps completed
`wrong_attempts_per_step`	Array of ints	1D Array of I32	Sequence of wrong attempts made per step
`is_fault_completed`	Boolean	Boolean	Always `true`

Payload 3: Fault Started Acknowledgment (`fault_started`)

```
{
  "event": "fault_started",
  "fault_code": 101,
  "total_steps": 14,
  "status": "started"
}
```

5. LabVIEW Native Parsing (Cluster Wiring)

> **Unflatten From JSON Wiring**
>
> 1. Drop an **Unflatten From JSON** node (*Functions Palette → Programming → String → Flatten/Unflatten String → Unflatten From JSON*).
> 2. Create a **Type Definition Cluster** with: `event` (String), `fault_code` (I32), `step_number` (I32), `wrong_attempts` (I32), `total_steps` (I32), `status` (String), `is_fault_completed` (Boolean).
> 3. Wire UDP incoming string into `JSON string`, and Cluster into `type and defaults`.
> 4. Unbundle by Name to route `step_number` and `wrong_attempts` directly to scorecard indicators.